Subtle Spectral Difference Discriminative Deep Metric Learning With Spectral Center Construction for Hyperspectral Target Detection

Dehui Zhu, Yuetian Lu, Ping Zhong, *Senior Member, IEEE*, Bo Du, *Senior Member, IEEE*, and Liangpei Zhang, *Fellow, IEEE*

***Abstract*—Detection of targets for hyperspectral images (HSIs) persists as a fundamental task in remote sensing image processing. Exploring the discriminative ability of deep Siamese networks to distinguish targets from backgrounds is the mainstream method for target detection (TD) in HSIs. Nevertheless, these methods enhance the discriminative ability of networks by learning the separability distance between the target and the overall backgrounds, where the backgrounds are considered as a single category. As a result, they may struggle to effectively suppress backgrounds with solely subtle spectral differences from the target, resulting in a limited separability performance, and the inability to accurately detect the targets. To alleviate this problem, we propose a novel subtle spectral difference discriminative deep metric learning (S^2D^3ML)-based target detector for HSIs in this work. The proposed S^2D^3ML constructs a deep metric learning framework embedded with a discriminative constraint to learn a deep metric feature space for addressing limited separability, in which the subtle feature differences between targets and different ground objects can be distinguished. In addition, we investigate a new multiblock sparse representation score-based strategy to obtain sufficient samples and spectral centers of backgrounds for training the S^2D^3ML framework. Finally, the detection of targets is executed within the learned metric space. A comprehensive suite of experiments is rigorously conducted on four benchmark datasets, and the results indicate that the S^2D^3ML achieves superior performance in HSIs TD.**

***Index Terms*—Deep metric learning, hyperspectral image (HIS), sample selection, target detection (TD).**

I. Introduction

IN THE domain of remote sensing, hyperspectral sensors play a pivotal role by capturing a scope of the electromagnetic spectrum, yielding a hyperspectral image (HSI). This imagery is distinguished by its fine-grained spectral resolution, which is manifested through a series of closely spaced and narrow spectral bands [1], [2], [3], [4], [5]. The high-density arrangement of these bands endows HSI with an extensive reservoir of spectral information, which is pivotal for discerning a diverse array of materials, even those that are closely similar, as the reflectance spectra exhibit material-specific variations [6], [7], [8], [9]. The distinctive spectral signatures of materials render HSI a potent tool for observing and analyzing surface materials, thereby garnering significant interest in the realms of detection, classification, and subpixel mapping [10], [11], [12], [13], [14]. Among them, HSI target detection (TD) has emerged as a pre-eminent area of study, driven by its manifold real-world applications [5], [15], [16], [17], [18].

As a technique for the identification of ground objects, TD in HSIs is dedicated to the detection and location of targets of interest within the observed scene [19], [20]. Unlike salient object detection [21], [22], [23], [24], [25], which often relies on bounding boxes to delineate and identify discrete objects such as airplanes, ships, and specific regions of interest, TD in HSI is more akin to pixel-level detection, framing it as a binary classification challenge that involves categorizing each pixel as either target or background [26], [27], [28]. However, it is important to note that due to the limitation of target pixels, traditional classification paradigms may not be entirely suitable for hyperspectral TD, as the overall classification error can significantly impact detection accuracy [29], [30]. As a supervised technique, HSI TD methods usually necessitate a priori knowledge of the target spectrum, typically one or several target spectra, to perform TD [31], [32].

The development of hyperspectral TD methods can be dated back to the 1980s, with early efforts focused on the identification of specific materials through their distinctive spectral signatures [33]. Since then, the field has witnessed the development of a multitude of HSI TD techniques, which can be broadly categorized into three primary families, each with its own set of theoretical underpinnings.

A. Classical Hypothesis Model-Based Detectors

The genesis of hyperspectral TD can be attributed to the early development of hypothesis model-based methods. This

Received 17 July 2025; revised 10 November 2025; accepted 24 November 2025. Date of publication 1 December 2025; date of current version 8 December 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 42501477, in part by China National Postdoctoral Program for Innovative Talents under Grant BX20240492, and in part by China Postdoctoral Science Foundation under Grant 2025M774409. *(Corresponding authors: Dehui Zhu; Ping Zhong.)*

Dehui Zhu, Yuetian Lu, and Ping Zhong are with the National Key Laboratory of Automatic Target Recognition, College of Electrical Science and Technology, National University of Defense Technology, Changsha 410073, China (e-mail: dhzhu95@163.com; yuetianlu19@nudt.edu.cn; zhongping@nudt.edu.cn).

Bo Du is with the National Engineering Research Center for Multimedia Software, Institute of Artificial Intelligence, School of Computer Science, Wuhan University, Wuhan 430079, China (e-mail: gunspace@163.com).

Liangpei Zhang is with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China (e-mail: zlp62@whu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3639103

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:13:29 UTC from IEEE Xplore. Restrictions apply.

category includes a series of techniques structured around distinct background assumptions. These include the structured background methods, exemplified by the CEM [34] and the OSP [35]; the unstructured background methods, such as the AMF [36] and the ACE [37]; and the hybrid background models, represented by the HUD [38] and the HSD [38]. These detectors are predicated on specific statistical hypotheses regarding the distribution of pixels within HSIs. However, the effectiveness of these methods is often impeded by a critical discrepancy. It is often observed that the theoretical hypotheses regarding the distribution of samples within HSIs do not align with practical scenarios, leading to limited outcomes in detection performance. To alleviate this problem, some current works [39] have attempted to modify the detector based on hypothetical models and have achieved decent results, such as hCEM [17], TCIMF [40], BA-TCIMF [41], TCIMBS [42], and some iterative-based and iterative random training sampling-based modified CEM method [43], [44]. Some other works construct detectors based on signal-to-noise ratio and spectral angle theories, and also achieve improved performance [45].

B. Machine Learning-Based Detectors

Recent years have witnessed the emergence of methods driven by machine learning techniques. These detectors are constructed based on different machine learning methods, including kernel methods [46], [47], sparse representation [48], [49], [50], [51], [52], metric learning [53], [54], and so on. Divergent from the hypothesis model-based detector predicated on specific assumptions, these machine learning-based approaches endeavor to identify superior metrics or suitable representations for disparate pixels, thereby facilitating the differentiation between targets and their backgrounds. For instance, representation-based detectors, such as STD [48] and CSCR [49], use the contextual pixels surrounding the middle one to represent one, thereby achieving an apt expression of each pixel within the imagery scene [55]. Kernel-based approaches, such as KSMF [46] and KTCIMF [56], on the other hand, translate raw data into a kernel feature space, seeking to improve the separability between targets and backgrounds through an improved measurement. Meanwhile, metric learning- based methods, such as MMML [53] and MIML [54], aspire to derive a precise measure of distance within a novel metric space, informed by the samples, to perform detection [57]. Moreover, low-rank and sparse matrix decomposition techniques are also used for target detector construction, where the backgrounds and target signal sources are in two separate subspaces, such as BA-OSP [58] and DSOSP [58]. To date, the exploration of machine learning-based target detectors in HSIs continues to be a vibrant area of research in HSI analysis.

C. Deep Learning-Based Detectors

The emergence of deep learning-based detectors has marked a paradigm shift in the domain of TD within HSIs, emerging as the predominant approach. These detectors utilize deep networks to capture both spatial and spectral nuances of the HSIs. This dual capability enables a precise demarcation between targets and backgrounds, thereby enhancing the detection performance. According to the presence or absence of supervision information during training, deep learning-based detectors can be roughly divided into two categories: target detectors in a supervised manner and target detectors in an unsupervised manner. Detectors in the first category usually select or synthesize pixel samples and then form an end-to-end detector by constructing pixel pairs to distinguish targets and backgrounds. Notable examples of such methods include CNND [59], HTD-Net [60], TSCNTD [61], STTD [62], SASN [63], and TSTTD [64]. Conversely, another type of method usually employs deep autoencoders to learn and extract discriminative features. These features then serve as the basis for TD, exemplified by methods such as SRUN [65], ULMMDL [33], HTD-IRN [66], ABLAL [67], and GSSTD [68]. Furthermore, some advanced deep learning mechanisms are employed in HSI TD, such as deep metric learning [69], [70], transfer learning [71], [72], and transformer [73], [74].

The advanced deep learning-based detectors spare no effort to capture nonlinear and deep features for detection, denoting more robust detection performance for HSIs TD. In essence, most of them are learning a suitable metric distance to distinguish the target from the backgrounds, where the distance between the prior spectrum and the targets is narrow, and the distance between the prior spectrum and the backgrounds is stretched. However, these detectors primarily enhance the discriminative ability of the network by increasing the separability distance between the target and the overall backgrounds when designing detectors. They consider the background as a single category, which makes it difficult to effectively suppress the backgrounds with small spectral differences from the target, resulting in limited separability and the inability to separate the target from the backgrounds.

To address this limitation, this study presents a novel subtle spectral difference discriminative deep metric learning (S^2D^3ML)-based target detector for HSIs. In S^2D^3ML, a subtle spectral difference discriminative constraint (S^2D^2C) is embedded within the deep metric network, where the similarities between diverse backgrounds and the target are performed separately. In this case, the subtle feature differences between diverse ground objects can be distinguished, thereby enabling effective background suppression and contributing to competitive detection performance. The main contributions of this study are outlined in the following.

1) To ensure the acquisition of ample background samples and accurate spectral centers across varied backgrounds, we propose a new multiblock sparse representation-based strategy for the selection of background pixels and the learning of spectral centers. It ensures the purity of the background samples and mitigates contamination, thereby facilitating the selection of a robust set of background samples and the extraction of representative and precise spectral features for diverse backgrounds.
2) A novel target detector named S^2D^3ML is investigated for TD in HSIs. The proposed S^2D^3ML constructs a new deep metric framework with a well-designed subtle spectral difference constraint to learn a deep

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:13:29 UTC from IEEE Xplore. Restrictions apply.

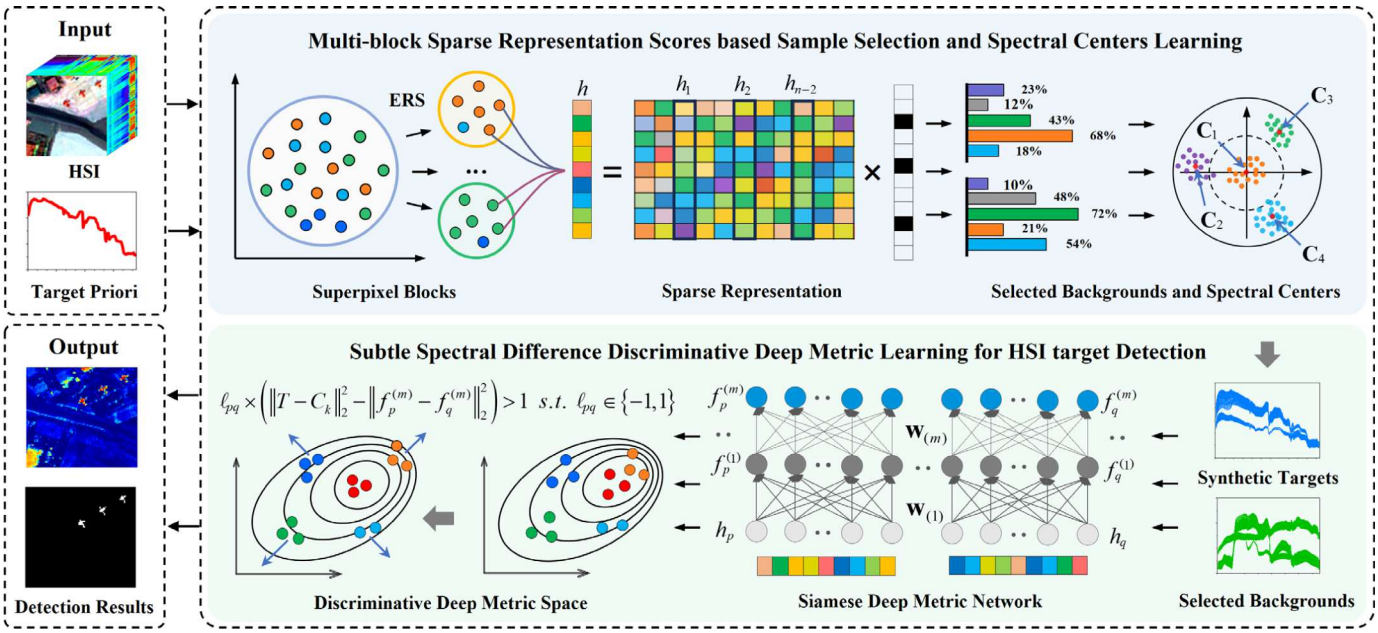


Fig. 1. Flowchart of the S²D³ML for TD in HSIs.

discriminative metric space, in which the subtle feature differences between targets and different ground objects can be distinguished, so as to address limited separability and enhance TD performance.

The sections of this article are meticulously organized as follows. First, Section II delves into the exposition of the proposed S²D³ML target detector, which commences with an overview of the multiblock sparse representation scores-based strategy and a comprehensive exposition of the S²D³ML framework. Second, Section III is dedicated to the delineation of experiments and analyses conducted on four benchmark HSIs. Finally, the conclusion of this study is drawn in Section IV.

II. Proposed S²D³ML Target Detector

In this section, we delve into an exhaustive exposition of the proposed S²D³ML target detector for HSIs. The schematic representation of the S²D³ML process is delineated in Fig. 1, highlighting its two principal constituents.

1) Initially, we introduce a multiblock sparse representation scores-driven strategy for background pixel selection and spectral center learning, providing background samples and spectral centers of each potential background category for the subsequent network training. Concurrently, the limited target sample is mitigated through the generation of subpixel targets using the prior target spectra and the preselected background samples.
2) Subsequently, we present a Siamese metric framework, which is augmented with a subtle spectral difference constraint, facilitating the learning of a deep metric space. Within this space, the subtle feature differences between diverse ground objects can be distinguished, thereby amplifying the separability. Ultimately, the detection of targets is accomplished by applying a basic detector within the learned deep discriminative metric space.

A. Background Pixel Selection and Spectral Center Learning

In the realm of HSI, the task of TD is inherently complex due to the lack of a priori background information. Therefore, the effective selection of background pixels is crucial for enhancing the accuracy of TD since the precision of this step is directly linked to the fidelity of the network training. Besides, to help suppress backgrounds, learning typical background features, i.e., the spectral centers of each potential background category, is also a critical juncture for the designed deep metric network. In light of these requirements, we investigate an innovative multiblock sparse representation scores-based strategy for background pixel selection and spectral center learning.

The goal of the strategy is to obtain typical background samples from the image, and it needs to ensure the purity of the background, that is, no interference from targets or anomalies, and the category of backgrounds is complete. Sparse representation, as a widely used technique in HSIs, is usually utilized to represent signals using a small number of atoms (pixels). During sparse representation, a higher frequency score of the atom in a series of atoms means the atom performs a stronger representativeness. In addition, the scores of anomaly pixels or noise pixels will reduce as they are a very small percentage. These attributes would avoid contamination and ensure the purity and typicality of the backgrounds. Moreover, the selection of background samples in various blocks can maximize the use of all types of backgrounds, which makes the category of backgrounds more complete. To this end, the multiblock sparse representation is employed in this work.

Let $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_N] \in \mathbb{R}^{b \times N}$ represent the input HSI characterized by N pixels and b spectral bands. At the outset,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:13:29 UTC from IEEE Xplore. Restrictions apply.

our methodology employs a superpixel segmentation approach, specifically applied to the first principal component of the input image, to segment the scene into n superpixel blocks, which can be denoted as $\bar{\mathbf{H}}^i = \{\mathbf{h}_1^i, \mathbf{h}_2^i, \ldots, \mathbf{h}_{Ni}^i\}$, where $i = 1, 2, \ldots, n, \bar{\mathbf{H}}^i$ is the ith superpixel block and Ni is the number of pixels in this superpixel block. In this work, the superpixel segmentation technique of choice is the entropy rate superpixel segmentation, abbreviated as ERS [75], as it has better anti-noise performance and higher computational efficiency. The resultant n superpixel blocks serve as prospective blocks for the subsequent refinement of background pixel selection. Within these superpixel blocks, pixels in some blocks may exhibit a striking resemblance to the a priori target spectrum, thereby increasing the likelihood that pixels selected from these blocks are indeed target in nature. Consequently, the presence of such blocks necessitates their exclusion during the background pixel selection process. To address this, we leverage the sparse representation technique, which facilitates the reconstruction of the prior target spectrum utilizing the pixels within each superpixel block, as articulated in the following formulation:

$$\widehat{\alpha} = \arg\min \left\| \bar{\mathbf{H}}^i \alpha - T \right\|_2 \quad \text{s.t.} \|\alpha\|_0 \leq \sigma \tag{1}$$

where T symbolizes the priori target spectrum, α denotes the sparse vector, and σ represents the sparse level. To address the optimization problem, the orthogonal matching pursuit algorithm is employed, where the sparsity level is set to ten empirically. Through this method, we derive the reconstruction residuals for each superpixel block as follows:

$$r_i(T) = \left\| T - \bar{\mathbf{H}}^i \widehat{\alpha} \right\|_2, \quad i = 1, 2, \ldots, n. \tag{2}$$

It is obvious that blocks with small reconstruction residuals are prone to engender target-like samples, thereby confounding the background pixel selection process; hence, their exclusion is warranted. Utilizing empirical evidence, this work eliminates the first 5% of superpixel blocks that correspond to the smallest reconstruction residuals, thereby refining the selection process in subsequent stages.

For the remained superpixel blocks, our method deploys a multiblock sparse representation scores-based strategy to select background pixels. In particular, the procedure unfolds in the following sequence of steps.

1) Construct dictionaries for each pixel in superpixel blocks

$$\mathbf{D}\left(\mathbf{h}_k^i\right) = \left\{\mathbf{h}_1^i, \mathbf{h}_2^i, \ldots, \mathbf{h}_{k-1}^i, \mathbf{h}_{k+1}^i, \ldots, \mathbf{h}_{Ni}^i\right\}. \tag{3}$$

2) Calculate the sparse vectors of each pixel

$$\widehat{\theta}_k^i = \arg\min \left\| \mathbf{D}\left(\mathbf{h}_k^i\right) \theta_k^i - \mathbf{h}_k^i \right\|_2 \quad \text{s.t.} \left\| \theta_k^i \right\|_0 \leq \sigma. \tag{4}$$

3) Count the frequency of each pixel in superpixel blocks

$$f\left(\mathbf{h}_k^i\right) = \sum_{r=1}^{Ni-1} L\left(\widehat{\theta}_r^i\right), \quad r \neq k \tag{5}$$

where $L\left(\widehat{\theta}\right) \in \{0, 1\}$ is the element in the sparse vector.

4) Calculate the sparse representation scores of each pixel

$$R\left(\mathbf{h}_k^i\right) = \frac{f\left(\mathbf{h}_k^i\right)}{Ni} \times 100\%. \tag{6}$$

5) Select the pixel with the largest score as the background sample in each superpixel block.

Upon the calculation of the aforementioned steps, a substantial ensemble of background samples embodying typical background characteristics is procured. Furthermore, during the computation of the sparse representation score, the presence of anomalies or noise within the diverse regions of superpixel blocks yields diminished scores. This attribute ensures the purity of the derived background samples and avoids contamination. In this work, for each superpixel block, a typical background pixel is selected as a sample for subsequent procedures. Following the elimination of the first 5% of superpixel blocks predicated on their diminutive reconstruction residuals, the tally of selected samples is determined by the formula $Nb = \text{floor}\ (0.95^* n)$, where the floor($x$) connotes the operation of rounding down to the nearest integer.

Subsequent to the selection of background pixels, there arises the imperative to learn the typical background features for each potential background class. This discernment is instrumental in facilitating the suppression of backgrounds during the induction of deep metric space learning. In light of the indeterminate number of background categories, this study employs the mean shift clustering (MSC) algorithm [76] to categorize the selected background samples. The mean shift algorithm is favored for its nonreliance on a predefined category parameter and its proficiency in delineating classes of arbitrary geometrical configurations. The clustering process, predicated on the selected background samples, yields the spectral centers of various categories, denoted as C_k ($k \geq 0$), representing the spectral center of the kth background category when $k \neq 0$. Concurrently, the background samples are apportioned into their respective classes and aligned with each spectral center, thereby expediting the subsequent network training regimen.

Similar to the case with background samples, the availability of prior target samples for deep learning networks is also constrained, typically offering one target spectrum within the HSI task. To address the limitation of target samples, this work draws upon a straightforward yet effective method for the synthesis of subpixel target samples, as elucidated in [61], where the selected background samples are implanted into the target prior spectrum in different proportions to synthesize the target sample. For convenience, each background sample is integrated into the target spectrum with an arbitrary ratio that oscillates between 0% and 20%, thereby generating a pseudo target sample (see the gray arrow on the right of Fig. 1). Consequently, Nt target samples can be generated, and apparently, the number of generated target samples is the same as that of background samples.

B. Discriminative Deep Metric Learning Framework

As previously mentioned, TD aims to label pixels as target or background, which can also be considered as a binary classification task. In metric learning theory, a distance metric can be defined between any two samples, and it is often assumed that samples with a close distance belong to the same category. If a suitable metric distance can be found, an accurate space

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:13:29 UTC from IEEE Xplore. Restrictions apply.

can be established to describe the distance between samples, maximizing the separation of target samples and background samples. In addition, the powerful nonlinear feature learning capabilities of deep networks can provide advanced feature representations of hyperspectral data, particularly continuous spectral information, for establishing suitable distance metrics. These aspects motivate us to extend the deep metric learning to hyperspectral TD.

To establish a suitable metric distance to separate targets from backgrounds, a S²D³ML framework is presented as follows.

First of all, pixel pairs for network training are constructed separately utilizing the prior target spectra, generated target samples, and selected background samples, collectively denoted as $(\mathbf{h}_p, \mathbf{h}_q)$, with an assigned label $\ell \in \{-1, 1\}$. It is important to highlight that a pixel pair labeled $\ell = 1$ constitutes a positive sample, formed by pairing the prior target spectra with a pixel from the generated target samples. Conversely, a pixel pair labeled $\ell = -1$ is identified as a negative sample, consisting of the prior target spectra paired with a pixel from the background samples. To facilitate the learning of a discriminative deep metric feature space for TD, the individual pixel pairs are input as spectral vectors into the designed Siamese deep metric network, as illustrated in Fig. 1. The two subnetworks with the same structure in the Siamese network share weights and are trained simultaneously, which can ensure that the same type of input spectral vector is mapped to a unified feature space, facilitating fair comparison and the discovery of spectral differences between two input pixel pairs.

For a pixel pair $(\mathbf{h}_p, \mathbf{h}_q) \in \mathbb{R}^b$, the output representations following feature extraction in the spectral dimension of the first layer of the Siamese metric network can be expressed as follows:

$$f_p^{(1)} = g\left(\mathbf{W}_{(1)}\mathbf{h}_p + \mathbf{b}_{(1)}\right) \tag{7}$$

$$f_q^{(1)} = g\left(\mathbf{W}_{(1)}\mathbf{h}_q + \mathbf{b}_{(1)}\right) \tag{8}$$

where $g(\cdot)$ is the activation function, and $\tanh(\cdot)$ is selected in this work, $\mathbf{W}_{(1)} \in \mathbb{R}^{N_{(1)} \times b}$ represents the nonlinear mapping weight matrix of the first layer, $\mathbf{b}_{(1)} \in \mathbb{R}^{N_{(1)}}$ is the bias vector, and $N_{(1)}$ indicates the number of nodes in this layer. Given that the network is designed as a Siamese deep metric network, it should be noted that the two spectral vectors utilize the same weights, as represented in (7) and (8). The number of bands in the input spectral vector corresponds to b; therefore, the number of nodes in this layer is set to $N_{(1)} = b$. Subsequently, the output representations are passed through the next network layer to extract further useful features, with the output represented by the following equations:

$$f_p^{(2)} = g\left(\mathbf{W}_{(2)}f_p^{(1)} + \mathbf{b}_{(2)}\right) \tag{9}$$

$$f_q^{(2)} = g\left(\mathbf{W}_{(2)}f_q^{(1)} + \mathbf{b}_{(2)}\right). \tag{10}$$

In the same way, the outputs for the mth layer of the Siamese deep network can be derived and are represented by the following:

$$f_p^{(m)} = g\left(\mathbf{W}_{(m)}f_p^{(m-1)} + \mathbf{b}_{(m)}\right) \tag{11}$$

$$f_q^{(m)} = g\left(\mathbf{W}_{(m)}f_q^{(m-1)} + \mathbf{b}_{(m)}\right) \tag{12}$$

where $\mathbf{W}_{(m)} \in \mathbb{R}^{N_{(m)} \times b}$ denotes the nonlinear mapping weight matrix for the mth layer, $f_p^{(0)} = \mathbf{h}_p$ and $f_q^{(0)} = \mathbf{h}_q$. To ensure the spectral vectors maintain continuity, the number of nodes $N_{(m)}$ in each layer is uniformly set to match the number of spectral bands, i.e., $N_{(m)} = b$, in this study. For a network comprising m layers, the distance between the two pixels in a pixel pair $(\mathbf{h}_p, \mathbf{h}_q)$ can be computed using the formulation

$$d^2\left(\mathbf{h}_p, \mathbf{h}_q\right) = \left\| f_p^{(m)} - f_q^{(m)} \right\|_2^2. \tag{13}$$

In general, after the learning process of the deep metric network, the distance between positive pixel pairs should be smaller than that between negative sample pairs. A common approach to achieve this is by enforcing a predefined threshold within the model, such that the distance between positive pixel pairs remains below this threshold, while the distance between negative pixel pairs exceeds it. For instance, in [77], a fixed threshold δ is set to ensure a margin distance between positive and negative pixel pairs, as formulated by the following equation:

$$\ell_{pq} \times \left(\delta - d^2\left(\mathbf{h}_p, \mathbf{h}_q\right)\right) > 1 \quad \text{s.t.} \ell_{pq} \in \{-1, 1\}. \tag{14}$$

However, in (14), the threshold value is fixed and designed to capture a specific margin distance. This implies that the model treats the background sample used to construct the negative pixel pair as a homogeneous category. Nevertheless, the negative pixel pairs consist of various background categories, some of which exhibit extremely subtle spectral differences from the target. In such cases, the model may struggle to effectively suppress backgrounds with small spectral differences from the target, leading to limited separability performance and challenges in accurately distinguishing the target from the background.

Therefore, alterable margin distances should be considered to better capture the differences between various backgrounds and the target a priori. This would facilitate the learning of a more suitable and discriminative deep metric space, thereby enhancing background suppression and separability. Inspired by this, we propose a subtle S²D²C where different similarities between the target priori spectrum and background serve as the threshold, formulated as

$$\begin{aligned} &\ell_{pq} \times \left(\|T - C_k\|_2^2 - \left\| f_p^{(m)} - f_q^{(m)} \right\|_2^2\right) > 1 \\ &\text{s.t.} \quad \ell_{pq} \in \{-1, 1\}, \quad k = 0, 1, \ldots, K \end{aligned} \tag{15}$$

where K is the number of background clusters via MSC, k value corresponds to the sample category of the constructed pixel pair, $k = 0$ corresponds to a positive pixel pair, and $k \neq 0$ refers to a negative pixel pair.

As observed from (15), the margin distance between two pixels in a positive pixel pair is designed to be narrow, whereas the margin distance between two pixels in a negative pixel pair is generally broader. In addition, for background samples in negative pixel pairs, the margin distances are adjustable, reflecting their varying spectral differences from the target a priori. Thus, the deep network can learn a more suitable and discriminative deep metric feature space, which contributes to effective background suppression and improved separability.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:13:29 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPREHENSIVE SPECIFICATION OF FOUR HSI DATASETS

Dataset	Number of Bands	Sensor	Image Size	Pseudo Color RGB	Spectral Resolution	Spatial Resolution	Target	Target Proportion
Sandiego	189	AVIRIS	200×200	56, 36, 18	10nm	3.5m/pixel	airplane	0.335%
RIT Campus	360	ProSpecTIR	200×200	55, 36, 18	5nm	1m/pixel	red panel	0.200%
MUUFL	64	-	325×220	-	10nm	<1m/pixel	green panel	0.091%
Reno Urban	356	ProSpecTIR	250×100	18, 35, 61	-	< 2m/pixel	car	0.856%

Algorithm 1 S^2D^3ML Target Detector for HSIs

Input: Hyperspectral image $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_N]$, priori target spectrum T, and number of subpixels n.

Output: Two-dimensional map of detection result.

Sample Selection and Spectral Center Learning:

1. Apply ERS to segment HSI into n superpixel blocks.
2. Construct dictionaries for each pixel via Eq.(3).
3. Calculate sparse vectors of each pixel by Eq.(4).
4. Count the frequency of each pixel by Eq.(5).
5. Calculate sparse representation scores via Eq.(6).
6. Select pixels with large scores as background samples.
7. Remove five percent samples for purification.
8. Learn spectral centers of diverse backgrounds via MSC.
9. Generate target samples in a synthetic way.

Discriminative Deep Metric Learning:

10. Build pixel pairs via target and background samples.
11. Compute the distance between pixel pairs by Eq.(13)
12. Apply the discriminative constraint via Eq.(15).
13. Train and optimize the deep Siamese metric network.
14. Obtain the learned deep discriminative metric space.

Target Detection:

15. Project test data into the discriminative metric space.
16. Perform detection of targets in a learned metric space.

The loss function for the deep network can be expressed as

$$\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n} \log \times \left(1 + \exp\left(1 - \ell_{pq} \times \left(\|T - C_k\|_2^2 - d^2\left(\mathbf{h}_p, \mathbf{h}_q\right)\right)\right)\right). \quad (16)$$

To train and optimize the network, we employ the adaptive moment estimation (Adam) algorithm in this work.

Upon optimization, we can obtain the projection matrix $\mathbf{M}$ from the optimized network. For any given input $\mathbf{h}_i \in \mathbb{R}^{b\times N}$, the transformed data can be obtained by the following equation:

$$\bar{\mathbf{h}} = \mathbf{M}^T\mathbf{h}. \quad (17)$$

Finally, we can detect targets via a basic detector in the deep discriminative metric space.

In summary, the proposed multiblock sparse representation scores-based strategy exploits the representation frequency of each pixel in superpixel blocks in the sparse representation operation to pick out typical background pixels. On the one hand, it emphasizes the contributions of different pixels in superpixel blocks, so as to obtain typical background features. On the other hand, it avoids potential anomalies or noise contaminations, which finally contribute to accurate and pure background sample selection. In addition, the S^2D^3ML especially constructs an S^2D^3ML framework, in which a subtle S^2D^2C is investigated to achieve a more suitable metric distance. As a result, the S^2D^3ML can distinguish subtle feature differences between targets and different ground objects, and address limited separability to enhance detection performance. The detailed procedure of the proposed S^2D^3ML framework is outlined in Algorithm 1.

III. EXPERIMENTS AND ANALYSIS

A. Hyperspectral Dataset Description

To rigorously evaluate the effectiveness of the S^2D^3ML method, an extensive suite of empirical experiments and analytical assessments has been conducted on four benchmark and real-world HSIs. These include the Sandiego dataset [78], the RIT Campus dataset [79], the MUUFL dataset [80], and the Reno Urban dataset [81]. A visual representation of the pseudocolor imagery and its corresponding ground truth for these datasets is delineated in Fig. 2. In the case of the Sandiego dataset, the target entities for detection are three airplanes, encompassing 134 pixels in total. This dataset is accessible from http://aviris.jpl.nasa.gov/data/free_data.html. The RIT Campus dataset covers several red planes with a total of 80 pixels designated as targets, captured within the campus of the Rochester Institute of Technology. This dataset is a valuable resource for HSI TD and is accessible at https://www.rit.edu/dirs/spectir-hyperspectral-airborne-2012. The MUUFL dataset, the third in our series, stands as a typical example of HSI TD datasets. It is available from the website https://github.com/GatorSense/MUUFLGulfport. Within this dataset, four planes with a distinct color are usually posited as targets for detection. In this work, the green plane, comprising 65 pixels, has been singled out as the target for detection. The final dataset is the Reno Urban dataset, which is real-world imagery that can be downloaded from SpecTIR's resource portal http://www.spectir.com/free-data-samples/. This dataset, captured by a ProSpecTIR sensor, presents 214 target pixels spread across a multitude of vehicles, offering a rich tapestry for analysis. An elaborate exposition of the four benchmark datasets is encapsulated in Table I, providing a comprehensive overview of their characteristics and relevance to the validation of the S^2D^3ML method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:13:29 UTC from IEEE Xplore. Restrictions apply.

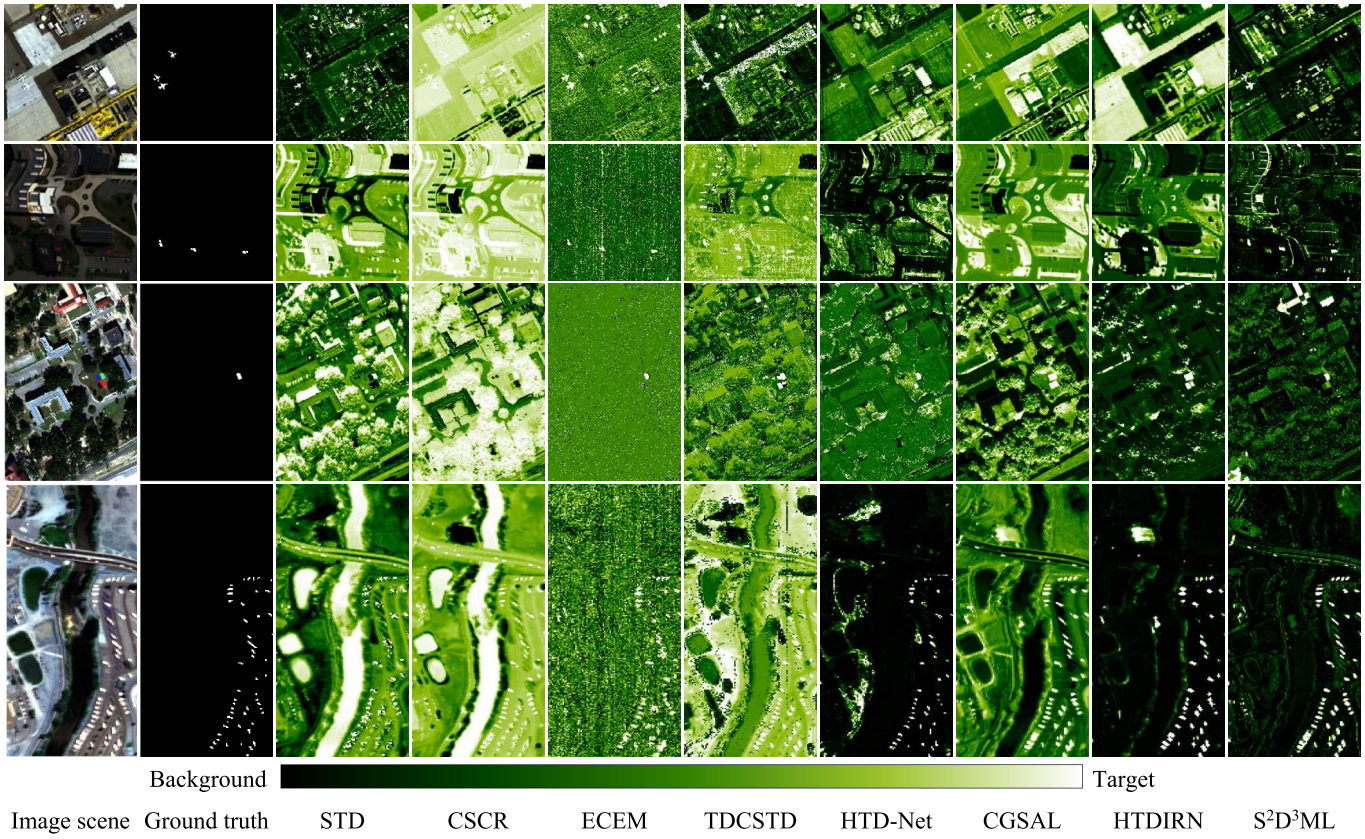


Fig. 2. Depth detection map comparison for four HSIs. (Top to bottom) Sandiego, RIT Campus, MUUFL, and Reno Urban.

B. Experimental Setup

This article undertakes a comprehensive assessment of the detection performance of the proposed S^{2}D^{3}ML framework, employing two widely used evaluative metrics, including the ROC curve and the AUC value.

1) ROC Curve: The 3-D ROC curve [82] is used in this article, which is generated as a function of probability of detection (PD), false alarm rate (FAR), and threshold τ, by introducing a specific threshold parameter τ as a third independent variable [82]. Three 2-D ROC curves can be achieved from the 3-D ROC curves, namely the 2-D ROC curve of PD-FAR, the 2-D ROC curve of PD-τ, and the 2-D ROC curve of FAR-τ. Among the three 2-D ROC curves, the 2-D ROC curve of PD-FAR is utilized to evaluate the effectiveness of detectors, and the 2-D ROC curve of PD-τ is used to evaluate the target detectability of detectors, while the 2-D ROC curve of FAR-τ is used to evaluate the background suppression ability of detectors.

2) AUC Value: The AUC value is explored in multiple dimensions, initially considering the area beneath the threshold-free ROC curve, denoted as $\mathrm{AUC_{PD\text{-}FAR}}$. Expanding upon this, we incorporate the 3-D ROC curve [82] and introduce two additional AUC indicators: $\mathrm{AUC_{PD\text{-}\backslash tau}}$ and $\mathrm{AUC_{FAR\text{-}\backslash tau}}$. Derived from these three AUC values, we further examine $\mathrm{AUC_{BS}}$ for background suppressibility, $\mathrm{AUC_{TD\text{-}BS}}$ for a composite measure of background suppressibility and target detectability, $\mathrm{AUC_{OD}}$ and $\mathrm{AUC_{ODP}}$ for overall detection accuracy and probability, and $\mathrm{AUC_{SNPR}}$ for signal-to-noise probability ratio, thereby providing a multifaceted evaluation of the detection results.

To substantiate the superiority of the S^{2}D^{3}ML method for TD in HSIs, its detection performance is compared with seven state-of-the-art target detectors. The comparators encompass STD [48], CSCR [49], ECEM [83], TDCSTD [20], HTD-Net [60], CGSAL [27], and HTD-IRN [66]. It is imperative to highlight that the priori target spectra are the same across all eight detectors to ensure equitable comparison. The tradeoff parameters for these comparative detectors have been meticulously adjusted, drawing from relevant literature, to ensure robust and reliable detection performance.

Regarding the configuration of parameters, a uniform sparse level of 10 is designated across all four datasets for the representation-based methods, STD, CSCR, and TDCSTD. This selection aligns with the sparsity level employed in the background pixel selection of the proposed S^{2}D^{3}ML method. The size of the dual window for STD and CSCR is adaptively tailored to correspond with the target size present within the imagery. Illustratively, in the Reno Urban dataset, the dual window sizes are configured as (5, 5) and (3, 3), whereas for the MUUFL dataset, they are escalated to (35, 35) and (17, 17) to encapsulate the respective target sizes. For the ECEM detector, the tradeoff parameters have been meticulously adjusted to ensure robust detection results, surpassing the performance limitations associated with default parameter values. As for the deep learning-based methods, i.e., HTD-Net and HTD-IRN, we adhere to the open-source code for experimental validation. Notably, for HTD-IRN, to uphold the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:13:29 UTC from IEEE Xplore. Restrictions apply.

fairness of comparisons, the given target spectrum is utilized as prior information, diverging from the default setup in the target spectrum generation module. The key parameters of the CGSAL detector are set as $\rho = 0.0001$ and $\mu = 0.01$ to ensure consistent detection effectiveness across all four HSIs. As for the proposed S^2D^3ML, we utilize the simple and widely used ACE detector to achieve detection results after learning distance metrics. The number of superpixel blocks is meticulously determined as $n = 300$ to select typical background samples. The bandwidth within the MSC method is set to 0.5 for the RIT Campus and MUUFL data and escalated to 1.0 for the remaining two HSIs, in which a flat kernel is employed for the method. The training regimen for the deep metric network within the S^2D^3ML framework is orchestrated with an epoch count of 50, a batch size of 128, and a learning rate of 0.001, respectively. These parameters are harmonized to cultivate a robust solution through the iterative optimization process.

C. Comparison of Detection Performance for Detectors

In this section, we delve into an exhaustive estimation of the detection performance of the S^2D^3ML method, compared against seven state-of-the-art detectors. This comparative analysis is articulated through robust metrics: the detection map, 3-D ROC curve, and AUC value.

First, we present a comparative analysis of the detection maps for the eight methods across four hyperspectral datasets, as depicted in Fig. 2, from a qualitative standpoint. It can be seen that the color of depth detection maps for the three representation-based methods is overall relatively light, indicating a poor background suppression performance. Similarly, the CGSAL and HTD-IRN also demonstrated poor background suppression performance on the Sandiego and RIT Campus datasets. For the HTD-Net, although it has a decent background suppression, there are still many cases of missed detections of targets, especially in the RIT Campus dataset.

Compared with other detectors, the depth map of the detection results of our proposed S^2D^3ML method is overall darker in color, indicating that most of the background can be well suppressed, while the real target position can be highlighted, enabling accurate target location and achieving good detection results. Across the four datasets, our proposed S^2D^3ML method consistently achieves superior detection performance in depth detection maps. This advantage stems from the subtle S^2D^2C embedded within the S^2D^3ML framework, which adeptly distinguishes subtle feature differences between diverse ground objects. Consequently, this enhances the separability between targets and backgrounds, thereby elevating the overall detection accuracy.

Second, a qualitative assessment of performance in TD through the 3-D ROC and the corresponding 2-D ROC curve analyses for the four HSIs is illustrated, as shown in Fig. 3. The distinct red curves in each subplot represent the ROC curve of our S^2D^3ML. Upon an examination of the second column in Fig. 4, it is discernible that the 2-D ROC curve of PD-FAR of the S^2D^3ML consistently maintains a front position relative to its counterparts across all four HSIs. This preeminence is indicative of a high PD and is particularly pronounced in the MUUFL and RIT Campus datasets, where the curve is positioned in the upper right corner of the plot. Similarly, in the 2-D ROC curve of PD-τ, the S^2D^3ML also achieves good results in both MUUFL and RIT Campus data. Although the S^2D^3ML does not obtain the best performance when it comes to the Sandiego and Reno Urban datasets for the 2-D ROC curve of PD-τ, low FAR under a small τ makes it achieve the best performance overall in the 2-D ROC curve of FAR-τ across the four datasets. For other detectors, the TDCSTD and HTD-Net exhibit constrained PD-FAR ROC performances within the RIT Campus dataset, while the representation-based methods do not perform well in the 2-D ROC curve of PD-τ since the FAR has not been effectively controlled due to the size of the double concentric window. Similarly, the HTD-Net demonstrates unsatisfactory outcomes in the Reno Urban scene for PD-FAR ROC and PD-τ ROC, characterized by a diminished PD and an inability to effectively suppress the FAR. The CGSAL method shows commendable PD-FAR ROC performance in the RIT Campus, MUUFL, and Reno Urban datasets, while falling short in the Sandiego dataset. As a spectral abundance learning-based detector, the CGSAL is sensitive to differences in spectral characteristics. The presence of metallic materials in the imagery that is similar to the airplane targets can confound the discernment of the CGSAL detector, leading to an impaired ability to suppress false alarms. In contrast, our proposed S^2D^3ML integrates a nuanced consideration of spectral centroids for each potential background category and constructs a subtle spectral difference discriminative deep metric feature space. This strategic formulation enhances the discriminative distance between targets and diverse backgrounds, thereby enabling effective background suppression and contributing to competitive 3-D ROC performance.

Moreover, we extend analyses to encompass the comparison of various AUC values for four datasets. This comparative evaluation is meticulously detailed in Tables II–V, with optimal values highlighted in bold and suboptimal values underscored. Each table enumerates eight AUC metrics for appraisal, namely $AUC_{PD\text{-}PF}$, $AUC_{PD\text{-}\backslash tau}$, $AUC_{PF\text{-}\backslash tau}$, AUC_{BS}, $AUC_{TD\text{-}BS}$, AUC_{ODP}, AUC_{OD}, and AUC_{SNPR}. Table II, which is dedicated to the RIT Campus dataset, reveals that our proposed S^2D^3ML detector secures the optimal AUC values across all indicators, thereby indicating its exceptional detection precision. The HTD-IRN method, while achieving suboptimal AUC values in all metrics, nonetheless exhibits superior detection performance. With the exception of the HTD-Net, the $AUC_{PD\text{-}PF}$ values of the remaining seven detectors surpass the threshold of 0.95. Notably, the TDCSTD, HTD-IRN, and the S^2D^3ML detector approach $AUC_{PD\text{-}PF}$ values nearing perfection, which indicates robust performance. In the context of the Reno Urban dataset, characterized by its complexity and the presence of numerous small targets, the S^2D^3ML method achieves the highest values in $AUC_{PD\text{-}FAR}$, AUC_{BS}, $AUC_{TD\text{-}BS}$, AUC_{ODP}, and AUC_{OD}. It also delivers the second-best AUC values in $AUC_{PF\text{-}\backslash tau}$ and AUC_{SNPR}. The CGSAL demonstrates decent performance for the Reno Urban data, where the complex background leads to suboptimal outcomes in several AUC indicators. Conversely, the HTD-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:13:29 UTC from IEEE Xplore. Restrictions apply.

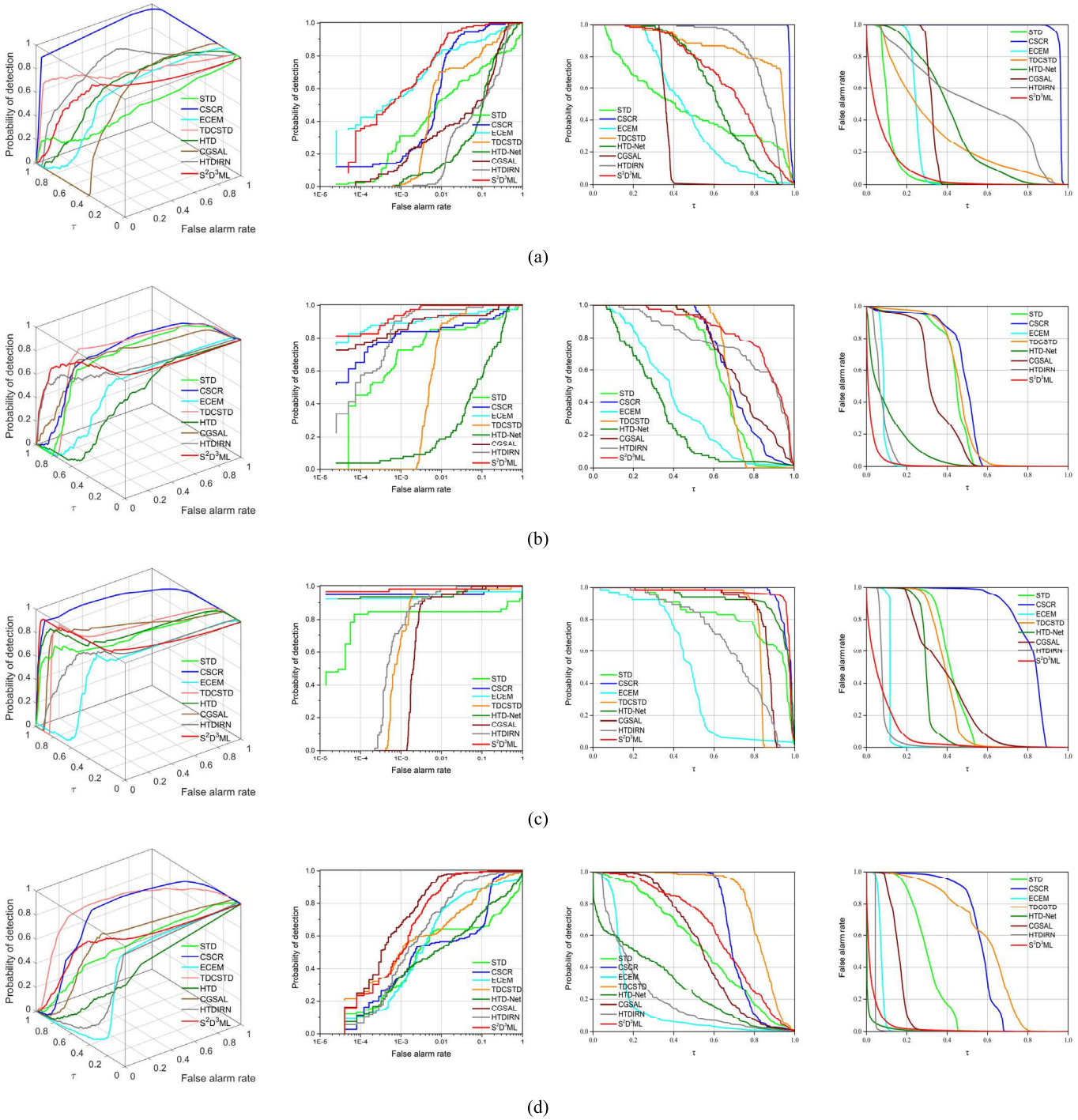


Fig. 3. Comparisons of 3-D ROC and corresponding 2-D ROC performance for four datasets. (a) Sandiego. (b) RIT Campus. (c) MUUFL. (d) Reno Urban. (Left to right) 3-D ROC curve, 2-D ROC curve of PD-FAR, 2-D ROC curve of PD-τ, and 2-D ROC curve of FAR-τ.

IRN underperforms in this scene, attributed to its reliance on binary classification loss for TD. This binary approach results in output values that are polarized, either tending toward 1 or 0, thereby increasing the likelihood of misclassifying background pixels as targets and leading to misclassifications and omissions. Tables IV and V further elucidate that our proposed S^{2}D^{3}ML detector attains optimal values in all AUC indicators, with the sole exception of $AUC_{PD-\backslash tau}$, where it exhibits a suboptimal value under the MUUFL dataset.

To sum up, the proposed S^{2}D^{3}ML method emerges as a formidable contender in the domain of HTD, demonstrating competitive performance across a series of evaluative metrics. These metrics include the detection maps, the 3-D ROC curves, and the AUC values. The S^{2}D^{3}ML leverages a multi-block sparse representation scores-based strategy for accurate background pixel selection. It further capitalizes on the subtle S^{2}D^{2}C within the deep metric network to effectively suppress diverse backgrounds. The synergistic approach endows the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:13:29 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISONS OF AUC VALUES IN DIFFERENT METHODS FOR THE RIT CAMPUS DATASET

Methods	$AUC_{PD\text{-}FAR}$	$AUC_{PD\text{-}\tau}$	$AUC_{FAR\text{-}\tau}$	AUC_{BS}	$AUC_{TD\text{-}BS}$	AUC_{ODP}	AUC_{OD}	AUC_{SNPR}
STD	0.9528	0.6459	0.4292	0.5236	0.2167	1.2167	1.1695	1.5049
CSCR	0.9677	0.7006	0.4729	0.4948	0.2278	1.2278	1.1955	1.4817
ECEM	0.9800	0.3970	0.0845	0.8955	0.3125	1.3125	1.2925	4.6971
TDCSTD	0.9921	0.6775	0.4519	0.5402	0.2257	1.2257	1.2178	1.4994
HTD-Net	0.8697	0.3138	0.1003	0.7694	0.2135	1.2135	1.0832	3.1282
CGSAL	0.9873	0.7322	0.3388	0.6485	0.3934	1.3934	1.3807	2.1613
HTDIRN	0.9979	0.7723	0.0762	0.9217	0.6961	1.6961	1.6940	10.1340
S^2D^3ML	**0.9998**	**0.8421**	**0.0217**	**0.9781**	**0.8204**	**1.8204**	**1.8202**	**38.8507**

TABLE III
COMPARISONS OF AUC VALUES IN DIFFERENT METHODS FOR THE RENO URBAN DATASET

Methods	$AUC_{PD\text{-}FAR}$	$AUC_{PD\text{-}\tau}$	$AUC_{FAR\text{-}\tau}$	AUC_{BS}	$AUC_{TD\text{-}BS}$	AUC_{ODP}	AUC_{OD}	AUC_{SNPR}
STD	0.8029	0.5441	0.3047	0.4982	0.2394	1.2394	1.0423	1.7857
CSCR	0.9350	0.7086	0.5640	0.3710	0.1446	1.1446	1.0796	1.2564
ECEM	0.9284	0.1711	0.0734	0.8550	0.0978	1.0978	1.0262	2.3326
TDCSTD	0.9501	**0.8275**	0.5808	0.3693	0.2466	1.2466	1.1967	1.4246
HTD-Net	0.8629	0.2787	**0.0106**	0.8523	0.2681	1.2681	1.1310	**26.3035**
CGSAL	0.9931	0.5449	0.1583	0.8348	0.3867	1.3867	1.3798	3.4433
HTDIRN	0.9799	0.2148	0.0450	0.9349	0.1698	1.1698	1.1497	4.7734
S^2D^3ML	**0.9960**	0.6416	0.0305	**0.9655**	**0.6112**	**1.6112**	**1.6072**	21.0473

TABLE IV
COMPARISONS OF AUC VALUES IN DIFFERENT METHODS FOR THE MUUFL DATASET

Methods	$AUC_{PD\text{-}FAR}$	$AUC_{PD\text{-}\tau}$	$AUC_{FAR\text{-}\tau}$	AUC_{BS}	$AUC_{TD\text{-}BS}$	AUC_{ODP}	AUC_{OD}	AUC_{SNPR}
STD	0.8886	0.8330	0.4209	0.4677	0.4122	1.4122	1.3008	1.9793
CSCR	0.9923	**0.9647**	0.8059	0.1864	0.1588	1.1588	1.1511	1.1970
ECEM	0.9688	0.4719	0.1164	0.8524	0.3555	1.3555	1.3243	4.0539
TDCSTD	0.9911	0.8202	0.3891	0.6020	0.4312	1.4312	1.4223	2.1082
HTD-Net	0.9971	0.9213	0.3001	0.6970	0.6211	1.6211	1.6182	3.0697
CGSAL	0.9949	0.8547	0.3837	0.6112	0.4710	1.4710	1.4659	2.2274
HTDIRN	0.9986	0.6986	0.0846	0.9140	0.6140	1.6140	1.6126	8.2583
S^2D^3ML	**0.9992**	0.9557	**0.0746**	**0.9246**	**0.8811**	**1.8811**	**1.8803**	**12.8058**

S^2D^3ML method with the preeminent performance among the evaluated methods.

D. Analysis of the Discriminative Constraint

In this work, a subtle S^2D^2C within the deep metric network is presented to help suppress diverse backgrounds and address limited separability for enhancing detection performance. To validate the effectiveness of the S^2D^2C, we conduct a comparative analysis against various metric constraints on the Sandiego and Reno Urban datasets. Under identical experimental conditions, we evaluate the detection performance through the ROC curve, AUC value, and binary detection maps, comparing the outcomes of the baseline deep metric model [as defined in (14)] with those of the model augmented with the S^2D^2C [as defined in (15)]. The comparative results are depicted in Figs. 4–6. It can be seen in Fig. 4 that the ROC curves with the S^2D^2C metric are ahead of those with the basic metric, and the AUC values with the S^2D^2C metric also surpass those with the basic metric. From the detection map results in Figs. 5 and 6, we can see that in both datasets, the model with the S^2D^2C metric can detect more target pixels as well as better background suppression. This is due to the fact that the S^2D^2C can contribute to learning alterable margin distances of the model to distinguish subtle feature differences between targets and different backgrounds, and then improve separability. The results indicate that the proposed S^2D^2C constraint can help suppress backgrounds and detect targets accurately to achieve excellent detection performance, which verifies the effectiveness of the S^2D^2C in the S^2D^3ML.

E. Analysis of the Number of Superpixel Blocks

In background sample selection within the S^2D^3ML model, the quantity of superpixel blocks assumes a pivotal role, as it directly influences the number of pixel pairs utilized for network training. To elucidate the impact of the number of superpixel blocks on detection performance, a parametric analysis was meticulously executed across four distinct datasets, where the number of superpixel blocks is roughly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:13:29 UTC from IEEE Xplore. Restrictions apply.

TABLE V
COMPARISONS OF AUC VALUES IN DIFFERENT METHODS FOR THE SANDIEGO DATASET

Methods	AUC_{PD-FAR}	$AUC_{PD-\tau}$	$AUC_{FAR-\tau}$	AUC_{BS}	AUC_{TD-BS}	AUC_{ODP}	AUC_{OD}	AUC_{SNPR}
STD	0.8314	0.4842	0.1181	0.7133	0.3661	1.3661	1.1975	4.1003
CSCR	0.9841	**0.9776**	0.9606	0.0235	0.0170	1.0170	1.0011	1.0177
ECEM	0.9644	0.4701	0.2459	0.7185	0.2242	1.2242	1.1886	1.9119
TDCSTD	0.9398	0.8529	0.3238	0.6160	0.5291	1.5291	1.4689	2.6342
HTD-Net	0.8428	0.6240	0.4038	0.4390	0.2202	1.2202	1.0630	1.5452
CGSAL	0.8791	0.3600	0.3234	0.5557	0.0367	1.0367	0.9158	1.1134
HTDIRN	0.8421	0.8447	0.5264	0.3157	0.3183	1.3183	1.1604	1.6047
S^2D^3ML	**0.9940**	0.6907	**0.0776**	**0.9164**	**0.6131**	**1.6131**	**1.6071**	**8.8991**

TABLE VI
EXECUTION TIME (SECONDS) COMPARISON OF DIFFERENT METHODS FOR FOUR HSIS

Dataset	STD	CSCR	ECEM	TDCSTD	HTD-Net	CGSAL	HTDIRN	S^2D^3ML
Sandiego	32.886 ±0.123	125.351 ±0.736	7.417 ±0.216	43.792 ±0.534	151.388 ±0.649	21.227 ±0.316	367.047 ±1.316	16.524 ±0.223
RIT Campus	23.978 ±0.203	38.229 ±0.296	31.126 ±0.996	34.781 ±0.334	206.783 ±0.963	18.192 ±0.366	461.683 ±1.740	19.349 ±0.117
MUUFL	85.129 ±0.378	1278.987 ±3.247	3.152 ±0.017	95.474 ±0.779	125.302 ±1.109	12.035 ±0.031	531.194 ±1.305	16.483 ±0.188
Reno Urban	4.127 ±0.047	3.904 ±0.174	25.203 ±0.676	20.369 ±0.770	171.764 ±1.207	17.186 ±0.271	281.083 ±1.848	11.871 ±0.105

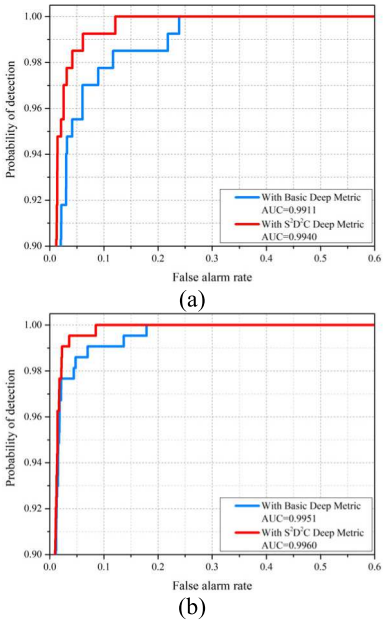


Fig. 4. 2-D ROC of PD-FAR comparison with different deep metric constraints for two HSI data. (a) Sandiego. (b) Reno Urban.

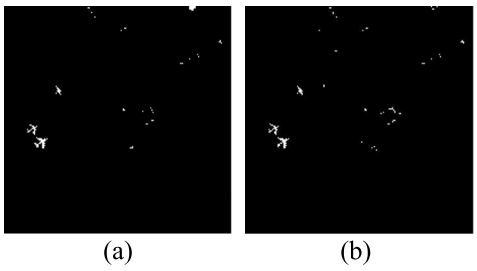


Fig. 5. Detection result comparison with different deep metric constraints for the Sandiego dataset: (a) with basic deep metric and (b) with S^2D^2C deep metric.

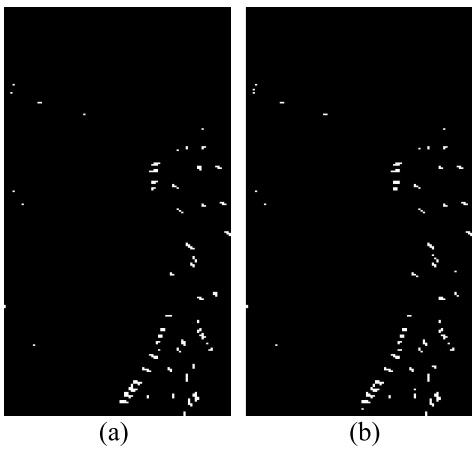


Fig. 6. Detection result comparison with different deep metric constraints for the Reno Urban dataset: (a) with basic deep metric and (b) with S^2D^2C deep metric.

set as [50, 100, 150, 200, 250, 300, 350, 400, 450, 500]. The results of AUC comparison with different numbers of superpixel blocks for four datasets are demonstrated in Fig. 7. As shown in Fig. 7, the AUC value exhibits an initial ascent with the augmentation of superpixel blocks, reaching a peak before experiencing a subsequent decline, particularly evident within the Sandiego and Reno Urban datasets. The reason is that a small number means a limited number of samples for training, which could result in an unstable solution for the designed network. Moreover, a large number of superpixel blocks could contribute to contamination during selection. For

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:13:29 UTC from IEEE Xplore. Restrictions apply.

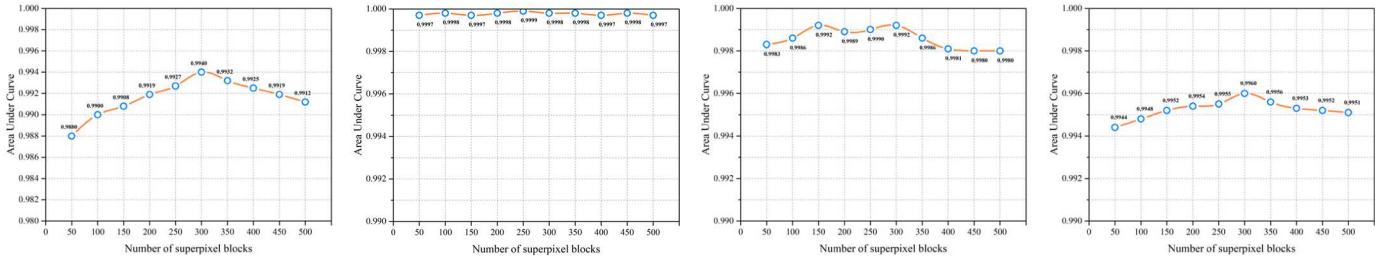


Fig. 7. AUC comparison with different numbers of superpixel blocks. (a) Sandiego. (b) RIT Campus. (c) MUUFL. (d) Reno Urban.

the four tested datasets, S^2D^3ML achieves better and more robust performance when the number of superpixel blocks is equal to 300. Therefore, the number of superpixel blocks set as 300 is reasonable for the S^2D^3ML detector to obtain superior detection results.

F. Analysis of the Execution Time

To analyze the computational complexity of the proposed method, an analysis of the execution time of different comparison detectors is presented, as shown in Table VI, where each method is repeated ten times under the same conditions to calculate the cost of time. For three representation-based detectors, i.e., STD, CSCR, and TDCSTD, the length of their running time is strongly related to the size of the dual window, and the running time is longer when the window size is larger, such as CSCR in Sandiego and MUUFL data, and STD and CSCR in MUUFL data. CSCR achieves the shortest execution time in the Reno Urban dataset compared with the other seven detectors, as the size of the dual window for the CSCR is small, i.e., (5, 5) and (3, 3). For ECEM, its computational cost is related to the number of layers; the running time is short when the number of layers is small. ECEM has the shortest time on both the Sandiego and MUUFL data among these detectors. For the iterative optimization algorithm CGSAL, the iteration convergence time determines the running time of the method. A running time of about 20 s guarantees a certain time efficiency for the CGSAL method when the image size is not very large. For two deep learning-based methods, i.e., HTD-Net and HTD-IRN, their execution time is generally longer than the other methods. The number of training pixel pairs of HTD-Net is huge, while the number of iterations of HTD-IRN is large, which is the main reason for their time-consuming nature. For our proposed S^2D^3ML, although it takes some time to construct the dictionary, the training process is much faster than other deep detectors. It takes a running time of about 15 s to complete the detection process, which is acceptable for TD in HSI.

IV. Conclusion

In this article, a novel deep metric learning-based detector denoted as S^2D^3ML is introduced for HSIs TD. To suppress diverse backgrounds and further address limited separability, a deep metric learning framework embedded with the subtle S^2D^2C is investigated to learn a deep metric feature space, in which the subtle feature differences between the target and backgrounds can be distinguished. Besides, a multiblock sparse representation score-based strategy is proposed to provide sufficient background samples and the spectral centers of various backgrounds for the designed deep metric learning framework to learn the separability ability. The S^2D^3ML performs the detection of targets in the learned discriminative metric space. Comprehensive detection results conducted on four real-world hyperspectral datasets, including detection maps, 3-D ROC, and various AUC values, confirm the competitive performance of the S^2D^3ML in TD for HSIs, outperforming benchmarks by several counterparts.

In the future, work will concentrate on deep metric networks to address nonlinear mixing and spectral variability problems in hyperspectral TD. In addition, a subpixel detection case will be considered.

References

[1] N. M. Nasrabadi, "Hyperspectral target detection: An overview of current and future challenges," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 34–44, Jan. 2014.

[2] X. X. Zhu et al., "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 8–36, Dec. 2017.

[3] B. Zhao et al., "Multiple attention network for spartina alterniflora segmentation using multitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5402915.

[4] J. Wang, M. Zhang, W. Li, and R. Tao, "A multistage information complementary fusion network based on flexible-mixup for HSI-X image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 12, pp. 17189–17201, Dec. 2024.

[5] A. F. H. Goetz, "Three decades of hyperspectral remote sensing of the Earth: A personal view," *Remote Sens. Environ.*, vol. 113, pp. S5–S16, Sep. 2009.

[6] B. Zhao et al., "Intermediate domain prototype contrastive adaptation for spartina alterniflora segmentation using multitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[7] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[8] D. A. Landgrebe, "Hyperspectral image data analysis," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 17–28, Jan. 2002.

[9] X. Ma et al., "A lightweight hybrid convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[10] B. Du, Y. Zhang, L. Zhang, and D. Tao, "Beyond the sparsity-based target detector: A hybrid sparsity and statistics-based detector for hyperspectral images," *IEEE Trans. Image Process.*, vol. 25, no. 11, pp. 5345–5357, Nov. 2016.

[11] W. Wang, X. Ma, L. Leng, Y. Wang, B. Liu, and J. Sun, "A hybrid CNN based on global reasoning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:13:29 UTC from IEEE Xplore. Restrictions apply.

[12] C. Shi and C.-M. Pun, "Multiscale superpixel-based hyperspectral image classification using recurrent neural networks with stacked autoencoders," *IEEE Trans. Multimedia*, vol. 22, no. 2, pp. 487–501, Feb. 2020.
[13] X. Ma et al., "An ultralightweight hybrid CNN based on redundancy removal for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.
[14] J. Wang, W. Li, Y. Wang, R. Tao, and Q. Du, "Representation-enhanced status replay network for multisource remote-sensing image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 15346–15358, Nov. 2024.
[15] M. T. Eismann, A. D. Stocker, and N. M. Nasrabadi, "Automated hyperspectral cueing for civilian search and rescue," *Proc. IEEE*, vol. 97, no. 6, pp. 1031–1055, Jun. 2009.
[16] K. C. Tiwari, M. K. Arora, and D. Singh, "An assessment of independent component analysis for detection of military targets from hyperspectral images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 13, no. 5, pp. 730–740, Oct. 2011.
[17] Z. Zou and Z. Shi, "Hierarchical suppression method for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 330–342, Jan. 2016.
[18] B. Lu, P. Dao, J. Liu, Y. He, and J. Shang, "Recent advances of hyperspectral imaging technology and applications in agriculture," *Remote Sens.*, vol. 12, no. 16, p. 2659, Aug. 2020.
[19] D. Manolakis and G. Shaw, "Detection algorithms for hyperspectral imaging applications," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 29–43, Jan. 2002.
[20] D. Zhu, B. Du, and L. Zhang, "Target dictionary construction-based sparse representation hyperspectral target detection methods," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 4, pp. 1254–1264, Apr. 2019.
[21] J. Han, G. Cheng, Z. Li, and D. Zhang, "A unified metric learning-based framework for co-saliency detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 28, no. 10, pp. 2473–2483, Oct. 2018.
[22] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.
[23] X. Wu, D. Hong, J. Tian, J. Chanussot, W. Li, and R. Tao, "ORSIm detector: A novel object detection framework in optical remote sensing imagery using spatial-frequency channel features," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 5146–5158, Jul. 2019.
[24] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.
[25] D. Zhang, J. Han, G. Cheng, and M.-H. Yang, "Weakly supervised object localization and detection: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 5866–5885, Sep. 2022.
[26] D. G. Manolakis, D. Marden, and G. A. Shaw, "Hyperspectral image processing for automatic target detection applications," *Lincoln Lab. J.*, vol. 14, no. 1, pp. 79–116, 2003.
[27] D. Zhu, B. Du, M. Hu, Y. Dong, and L. Zhang, "Collaborative-guided spectral abundance learning with bilinear mixing model for hyperspectral subpixel target detection," *Neural Netw.*, vol. 163, pp. 205–218, Jun. 2023.
[28] T. Guo, F. Luo, J. Guo, Y. Duan, X. Huang, and G. Shi, "Hyperspectral target detection with target prior augmentation and background suppression-based multidetector fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 1765–1780, 2024.
[29] Y. Dong, W. Shi, B. Du, X. Hu, and L. Zhang, "Asymmetric weighted logistic metric learning for hyperspectral target detection," *IEEE Trans. Cybern.*, vol. 52, no. 10, pp. 11093–11106, Oct. 2022.
[30] Z. Huang, L. Fang, and S. Li, "Subpixel-pixel-superpixel guided fusion for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 5998–6007, Sep. 2020.
[31] D. Manolakis, E. Truslow, M. Pieper, T. Cooley, and M. Brueggeman, "Detection algorithms in hyperspectral imaging systems: An overview of practical algorithms," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 24–33, Jan. 2014.
[32] A. W. Bitar, L.-F. Cheong, and J.-P. Ovarlez, "Sparse and low-rank matrix decomposition for automatic target detection in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5239–5251, Aug. 2019.
[33] Y. Li, Y. Shi, K. Wang, B. Xi, J. Li, and P. Gamba, "Target detection with unconstrained linear mixture model and hierarchical denoising autoencoder in hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 31, pp. 1418–1432, 2022.
[34] Q. Du, H. Ren, and C.-I. Chang, "A comparative study for orthogonal subspace projection and constrained energy minimization," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 6, pp. 1525–1529, Jun. 2003.
[35] C.-I. Chang, "Orthogonal subspace projection (OSP) revisited: A comprehensive study and analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 502–518, Mar. 2005.
[36] F. C. Robey, D. R. Fuhrmann, E. J. Kelly, and R. Nitzberg, "A CFAR adaptive matched filter detector," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 28, no. 1, pp. 208–216, Jan. 1992.
[37] S. Kraut and L. L. Scharf, "The CFAR adaptive subspace detector is a scale-invariant GLRT," *IEEE Trans. Signal Process.*, vol. 47, no. 9, pp. 2538–2541, Sep. 1999.
[38] J. Broadwater and R. Chellappa, "Hybrid detectors for subpixel targets," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 29, no. 11, pp. 1891–1903, Nov. 2007.
[39] C.-I. Chang, "Constrained energy minimization (CEM) for hyperspectral target detection: Theory and generalizations," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.
[40] H. Ren, "Target-constrained interference-minimized approach to subpixel target detection for hyperspectral images," *Opt. Eng.*, vol. 39, no. 12, pp. 3138–3145, Dec. 2000.
[41] J. Chen and C.-I. Chang, "Background-annihilated target-constrained interference-minimized filter (TCIMF) for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.
[42] X. Shang et al., "Target-constrained interference-minimized band selection for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6044–6064, Jul. 2021.
[43] B. Xue et al., "A subpixel target detection approach to hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5093–5114, Sep. 2017.
[44] C. Yu, B. Xue, M. Song, Y. Wang, S. Li, and C.-I. Chang, "Iterative target-constrained interference-minimized classifier for hyperspectral classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1095–1117, Apr. 2018.
[45] C.-I. Chang, "Hyperspectral target detection: Hypothesis testing, signal-to-noise ratio, and spectral angle theories," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.
[46] H. Kwon and N. M. Nasrabadi, "Hyperspectral target detection using kernel spectral matched filter," in *Proc. Conf. Comput. Vis. Pattern Recognit. Workshop*, Jul. 2004, p. 127.
[47] H. Kwon and N. M. Nasrabadi, "Kernel matched subspace detectors for hyperspectral target detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 28, no. 2, pp. 178–194, Feb. 2006.
[48] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Sparse representation for target detection in hyperspectral imagery," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 629–640, Jun. 2011.
[49] W. Li, Q. Du, and B. Zhang, "Combined sparse and collaborative representation for hyperspectral target detection," *Pattern Recognit.*, vol. 48, no. 12, pp. 3904–3916, Dec. 2015.
[50] X. Lu, W. Zhang, and X. Li, "A hybrid sparsity and distance-based discrimination detector for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1704–1717, Mar. 2018.
[51] X. Zhao, W. Li, C. Zhao, and R. Tao, "Hyperspectral target detection based on weighted Cauchy distance graph and local adaptive collaborative representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.
[52] R. Liu, J. Wu, D. Zhu, and B. Du, "Weighted discriminative collaborative competitive representation with global dictionary for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.
[53] Y. Dong, L. Zhang, L. Zhang, and B. Du, "Maximum margin metric learning based target detection for hyperspectral images," *ISPRS J. Photogramm. Remote Sens.*, vol. 108, pp. 138–150, Oct. 2015.
[54] B. Yang et al., "Multiple-instance metric learning network for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.
[55] Y. Zhang, B. Du, and L. Zhang, "A sparse representation-based binary hypothesis model for target detection in hyperspectral images,"

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:13:29 UTC from IEEE Xplore. Restrictions apply.

IEEE Trans. Geosci. Remote Sens., vol. 53, no. 3, pp. 1346–1354, Mar. 2015.

[56] T. Wang, B. Du, and L. Zhang, "A kernel-based target-constrained interference-minimized filter for hyperspectral sub-pixel target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 626–637, Apr. 2013.

[57] Y. Dong, Y. Zhang, and B. Du, "Symmetric information–theoretic metric learning for target detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1470–1480, 2022.

[58] C.-I. Chang and J. Chen, "Orthogonal subspace projection using data sphering and low-rank and sparse matrix decomposition for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8704–8722, Oct. 2021.

[59] W. Li, G. Wu, and Q. Du, "Transferred deep learning for hyperspectral target detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 5177–5180.

[60] G. Zhang, S. Zhao, W. Li, Q. Du, Q. Ran, and R. Tao, "HTD-Net: A deep convolutional neural network for target detection in hyperspectral imagery," *Remote Sens.*, vol. 12, no. 9, p. 1489, May 2020.

[61] D. Zhu, B. Du, and L. Zhang, "Two-stream convolutional networks for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6907–6921, Aug. 2021.

[62] W. Rao, L. Gao, Y. Qu, X. Sun, B. Zhang, and J. Chanussot, "Siamese transformer network for hyperspectral image target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526419.

[63] H. Gao, Y. Zhang, Z. Chen, F. Xu, D. Hong, and B. Zhang, "Hyperspectral target detection via spectral aggregation and separation network with target band random mask," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[64] J. Jiao, Z. Gong, and P. Zhong, "Triplet spectralwise transformer network for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[65] W. Xie, J. Yang, J. Lei, Y. Li, Q. Du, and G. He, "SRUN: Spectral regularized unsupervised networks for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1463–1474, Feb. 2020.

[66] D. Shen, X. Ma, W. Kong, J. Liu, J. Wang, and H. Wang, "Hyperspectral target detection based on interpretable representation network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[67] L. Sun, Z. Ma, and Y. Zhang, "ABLAL: Adaptive background latent space adversarial learning algorithm for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 411–427, 2024.

[68] X. Chen, Y. Zhang, Y. Dong, and B. Du, "Generative self-supervised learning with spectral–spatial masking for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[69] D. Zhu, B. Du, Y. Dong, and L. Zhang, "Target detection with spatial–spectral adaptive sample generation and deep metric learning for hyperspectral imagery," *IEEE Trans. Multimedia*, vol. 25, pp. 6538–6550, 2023.

[70] X. Xu, Z. Wang, C. Deng, H. Yuan, and S. Ji, "Towards improved and interpretable deep metric learning via attentive grouping," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 1189–1200, Jan. 2023.

[71] Y. Shi, H. Cui, Y. Yin, H. Song, Y. Li, and P. Gamba, "Transfer learning with nonlinear spectral synthesis for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[72] Y. Wu, Z. Li, B. Zhao, Y. Song, and B. Zhang, "Transfer learning of spatial features from high-resolution RGB images for large-scale and robust hyperspectral remote sensing target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[73] H. Qin, W. Xie, Y. Li, and Q. Du, "HTD-TS3: Weakly supervised hyperspectral target detection based on transformer via spectral–spatial similarity," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 16816–16830, Nov. 2023.

[74] Q. Yang, X. Wang, L. Chen, Y. Zhou, and S. Qiao, "CS-TTD: Triplet transformer for compressive hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[75] M.-Y. Liu, O. Tuzel, S. Ramalingam, and R. Chellappa, "Entropy rate superpixel segmentation," in *Proc. CVPR*, Jun. 2011, pp. 2097–2104.

[76] D. Comaniciu and P. Meer, "Mean shift: A robust approach toward feature space analysis," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 5, pp. 603–619, May 2002.

[77] J. Hu, J. Lu, and Y. Tan, "Discriminative deep metric learning for face verification in the wild," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 1875–1882.

[78] R. O. Green et al., "Imaging spectroscopy and the airborne visible/infrared imaging spectrometer (AVIRIS)," *Remote Sens. Environ.*, vol. 65, no. 3, pp. 227–248, Aug. 1998.

[79] A. Giannandrea et al., "The SHARE 2012 data campaign," *Proc. SPIE*, vol. 8743, May 2013, Art. no. 87430F.

[80] P. Gader, A. Zare, R. Close, J. Aitken, and G. Tuell, "MUUFL gulfport hyperspectral and LiDAR airborne dataset," University of Florida, Gainesville, FL, USA, Tech. Rep. REP-2013-570, Oct. 2013.

[81] Y. Liu, G. Gao, and Y. Gu, "Tensor matched subspace detector for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 4, pp. 1967–1974, Apr. 2017.

[82] C.-I. Chang, "An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5131–5153, Jun. 2021.

[83] R. Zhao, Z. Shi, Z. Zou, and Z. Zhang, "Ensemble-based cascaded constrained energy minimization for hyperspectral target detection," *Remote Sens.*, vol. 11, no. 11, p. 1310, Jun. 2019.



Dehui Zhu received the B.S. degree from China University of Petroleum, Qingdao, China, in 2018, and the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote sensing, Wuhan University, Wuhan, China, in 2023.

He is currently a Post-Doctoral Researcher with the National Key Laboratory of Automatic Target Recognition, College of Electrical Science and Technology, National University of Defense Technology, Changsha, China. His research interests include remote sensing image processing in machine learning and deep learning.



Yuetian Lu received the B.S. degree in information and communication engineering from the National University of Defense Technology (NUDT), Changsha, China, in 2022, where he is currently pursuing the Ph.D. degree with the National Key Laboratory of Science and Technology on ATR.

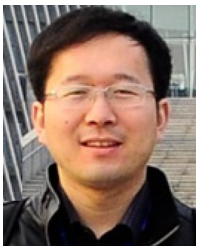
His research interests include hyperspectral image data and LiDAR data processing, computer vision, and deep learning.



Ping Zhong (Senior Member, IEEE) received the M.S. degree in applied mathematics and the Ph.D. degree in information and communication engineering from the National University of Defense Technology (NUDT), Changsha, China, in 2003 and 2008, respectively.

He is currently a Professor with the National Key Laboratory of Science and Technology on ATR, NUDT. His research interests include computer vision, machine learning, and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:13:29 UTC from IEEE Xplore. Restrictions apply.



Bo Du (Senior Member, IEEE) received the B.S. and Ph.D. degrees in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote sensing, Wuhan University, Wuhan, China, in 2005 and 2010, respectively.

He is currently a Professor with the School of Computer Science and the Institute of Artificial Intelligence, Wuhan University, where he is also the Director of the National Engineering Research Center for Multimedia Software. He has more than 80 research articles published in IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON CYBERNETICS, IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING; 14 of them are ESI hot papers or highly cited papers. His major research interests include machine learning, computer vision, and image processing.

Dr. Du regularly serves as a Senior PC Member for the International Joint Conferences on Artificial Intelligence (IJCAI) and AAAI. He won the Highly Cited Researcher in 2019\2020 by the Web of Science Group. He won the IEEE Geoscience and Remote Sensing Society 2020 Transactions Prize Paper Award. He won the IJCAI Distinguished Paper Prize, the IEEE Data Fusion Contest Champion, and the IEEE Workshop on Hyperspectral Image and Signal Processing Best Paper Award in 2018. He served as the Area Chair for ICPR. He serves as an Associate Editor for *Neural Networks*, *Pattern Recognition*, and *Neurocomputing*. He also serves as a reviewer of 20 Science Citation Index (SCI) magazines, including IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IEEE TRANSACTIONS ON CYBERNETICS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and IEEE GEOSCIENCE AND REMOTE SENSING LETTERS.



Liangpei Zhang (Fellow, IEEE) received the B.S. degree in physics from Hunan Normal University, Changsha, China, in 1982, the M.S. degree in optics from Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China, in 1988, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 1998.

He was the Head of the Remote Sensing Division, State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University. He is also a "Chang-Jiang Scholar" Chair Professor appointed by the Ministry of Education of China. He was a Principal Scientist of China State Key Basic Research Project from 2011 to 2016, appointed by the Ministry of National Science and Technology of China to lead the Remote Sensing Program in China. His research interests include hyperspectral remote sensing, high-resolution remote sensing, image processing, and artificial intelligence.

Dr. Zhang is a fellow of the Institution of Engineering and Technology (IET). He is currently serving as an Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:13:29 UTC from IEEE Xplore. Restrictions apply.